

3rd Infodengue-Mosqlimate Dengue Challenge (IMDC): 2026 Sprint for dengue fever forecasts for Brazil

The Infodengue-Mosqlimate Dengue Challenge (IMDC) is an initiative led by the Mosqlimate and Infodengue.

The objective of this 2026 sprint is **to promote training of predictive models and to develop high-quality ensemble forecast models for dengue in Brazil.**

The challenge involves four validation tests and one forecast target. The period of interest spans from the epidemiological week (EW) 41 of one year to EW 40 of the following year, aligning with the typical dengue season in Brazil.

Validation test 1. Predict the weekly number of dengue cases by state (UF) in the 2022-2023 season [EW 41 2022- EW40 2023], using data covering the period from EW 01 2010 to EW 25 2022;

Validation test 2. Predict the weekly number of dengue cases by state (UF) in the 2023-2024 season [EW 41 2023- EW40 2024], using data covering the period from EW 01 2010 to EW 25 2023;

Validation test 3: Predict the weekly number of dengue cases by state (UF) in the 2024-2025 season [EW 41 2024- EW40 2025], using data covering the period from EW 01 2010 to EW 25 2024;

Validation test 4. Predict the weekly number of dengue cases in Brazil, and by state (UF), in the 2025-2026 season [EW 41 2025- EW40 2026], using data covering the period from EW 01 2010 to EW 25 2025;

Forecast. Predict the weekly number of dengue cases in Brazil, and by state (UF), in the 2026-2027 season [EW 41 2026- EW40 2027], using data covering the period from EW 01 2010 to EW 25 2026;

This year, the competition included four forecasting challenges:

- **Mandatory Challenge – Dengue (State Level):** Forecast dengue cases at the state (UF) level for all Brazilian states, except Espírito Santo.
- **Optional Challenge 1 – Dengue (City Level):** Forecast dengue cases for 15 selected cities.
- **Optional Challenge 2 – Chikungunya (State Level):** Forecast chikungunya cases at the state (UF) level for all Brazilian states, except Espírito Santo.
- **Optional Challenge 3 – Chikungunya (City Level):** Forecast chikungunya cases for 10 selected cities.

This document presents the results for the **Mandatory Challenge (Dengue – State Level)**. The results for the optional challenges are available in the following files:

- [optional_challenge_1.md](#) – Dengue (City Level)
- [optional_challenge_2.md](#) – Chikungunya (State Level)
- [optional_challenge_3.md](#) – Chikungunya (City Level)

Teams and models

Team	Institution	Country	Repository	La
Global Health Resilience (GHR)	Barcelona Supercomputing Center	Spain	https://github.com/dieivillano/3rd_imdc_bsc_ghr	BS
Cornell-PEH	Cornell University	United States	https://mosqlimate.org/BentoLab-DiseaseDynamics/3rd_imdc_cornell_bentolab	CC
LNCC ARP26	Laboratório Nacional de Computação Científica	Brazil	https://github.com/pesquefGH/3rd_imdc_lncc_lncc_arp26_dengue	LN
LNCC SURGE	Laboratório Nacional de Computação Científica	Brazil	https://github.com/pesquefGH/3rd_imdc_lncc_surge_model26_dengue	LN
LNCC CLIDENGO	Laboratório Nacional de Computação Científica	Brazil	https://github.com/americocunhajr/3rd_imdc_lncc_clidengo26dengue	LN
Epidemáticos - Prophet	FGV EMap	Brazil	https://github.com/EzequielEBS/3rd_imdc_emap_epidemáticos_prophet	EM
Epidemáticos - Sarimax	FGV EMap	Brazil	https://github.com/EzequielEBS/3rd_imdc_emap_epidemáticos_sarimax_state	EM
Grupo Modelamiento de datos en Dengue	Universidad del Valle	Colombia	https://github.com/germanavila09/3rd_imdc_universidad_del_valle_grupo_modelamiento_datos_dengue	UN

Team	Institution	Country	Repository	La
InfraMIND	IFGW Unicamp and BIFI Universidad de Zaragoza	Brazil	https://github.com/InfraMIND-models/3rd_imdc_ifgw_inframind-proteus	IFC
ISI Dengue	ISI Foundation	Italy	https://github.com/mattiamazzoli/3rd_imdc_isi_isi-dengue	ISI
MARD	Fiocruz	Brazil	https://github.com/asgouveiaa/3rd_imdc_fiocruz_mard	FIC
Return of the Forecast	Centre for Epidemic Response & Innovation (CERI)	South Africa	https://github.com/CERI-KRISP/CERI_Dengue_Forecasting_2026	CE
RKI_ZKI_LSTM_GEO	RKI	Germany	https://github.com/DiogoParreira/3rd_imdc_rki_rki_zki_ph_lstm_geo	RK
RKI_ZKI_PH	RKI	Germany	https://github.com/DiogoParreira/3rd_imdc_rki_rki_zki_ph	RK
NOIS PUC Arbocaster	Pontifical Catholic University of Rio de Janeiro (PUC-Rio)	Brazil	https://github.com/Dududidicao99/3rd_imdc_pucrio_arbocaster	PU
Recogna	UNESP	Brazil	https://github.com/joel-da-silva-cavalcanti-filho/3rd_imdc_unesp_recogna	UN
XGBSillas	FGV-EMAP	Brazil	https://github.com/scrocha/3rd_imdc_emap_xgbsillas	EM
SAKHAL	FGV-EMAP	Brazil	https://github.com/marciomacielbastos/3rd_imdc_fgv_sakhal	EM
ZEROLAGS	FIOCRUZ	Brazil	https://github.com/Luizsrs/3rd_imdc_fiocruz_zerolags	FIC
Pattern-Blue	FGV-EMAP	Brazil	https://github.com/ZuilhoSe/3rd_imdc_fgv_pattern-blue	EM
Dengue Oracle	FGV-EMAP	Brazil	https://github.com/eduardocorrearaujo/3rd_imdc_emap_lstm	EM
Neural Earth	Purdue University	United States	https://github.com/kamrul28890/3rd_imdc_purdue_neuralearth	PU
BB model	PROCC	Brazil	https://github.com/lsbastos/3rd_imdc_procc_bb_model	PR
DS-OKSTATE-26	Oklahoma State University	United States	https://github.com/haridas-das/DS-OKSTATE-2026	DS
NUS CERM	National University of Singapore	Singapore	https://github.com/SungmokJung/3rd_imdc_nus_nus-cerm	NL
AFYA	Afya	Brazil	https://github.com/Ricafya/3rd_imdc_afya_ric	AF
4 Mosquiteiras	Federal University of São Paulo (UNIFESP)	Brazil	https://mosqlimate.org/blaiate/3rd_imdc_-unifesp_-4mosqueteiras-	UN

Ranking

To rank the models, we computed the ratio between each model's Weighted Interval Score (WIS) and the WIS of a baseline model. As the baseline, we used the **BB** model, which is described in the following publication: <https://www.sciencedirect.com/science/article/pii/S2468042725000739>.

For each validation period (v), the ratio was calculated as

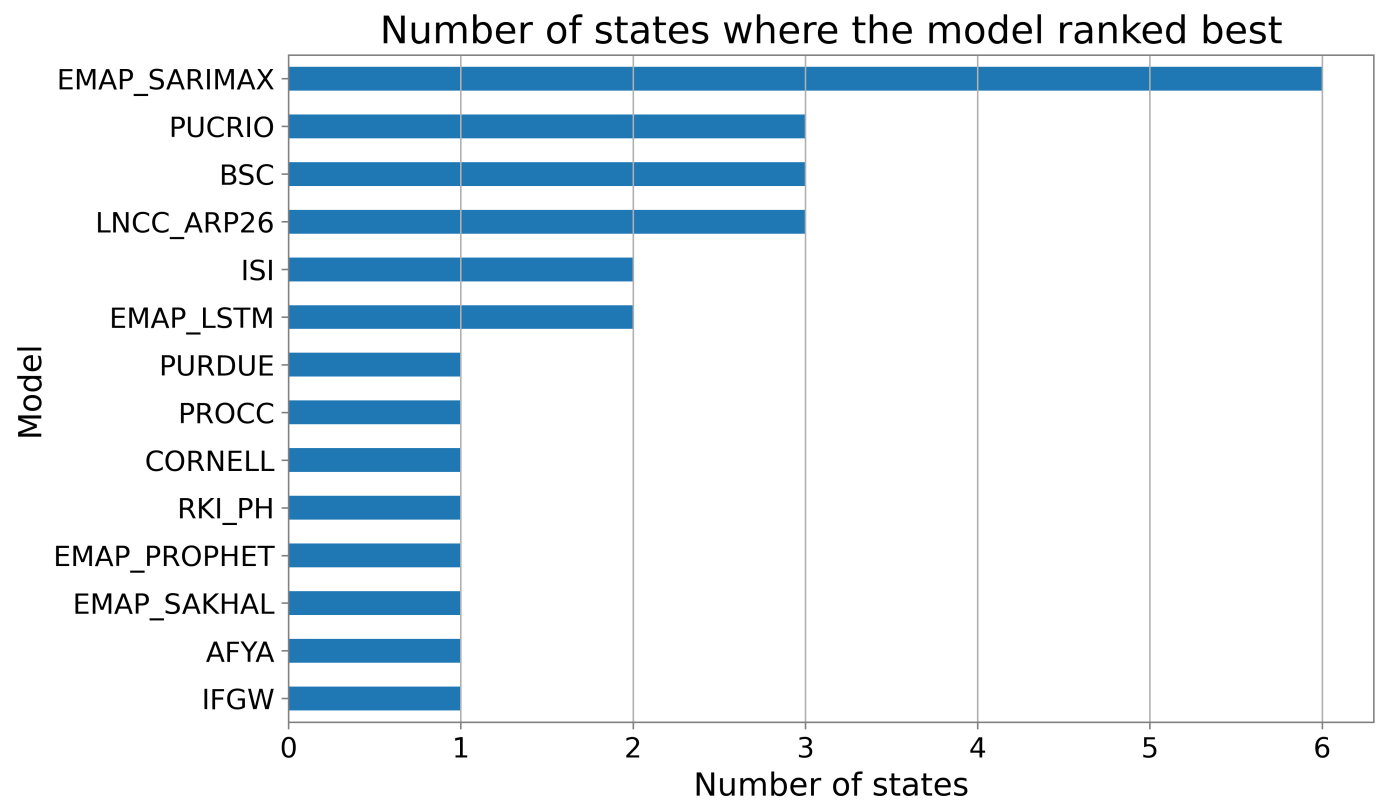
$$R_v = \frac{WIS(model, v)}{WIS(baseline, v)}$$

A value of ($R_v < 1$) indicates that the model outperformed the baseline, while ($R_v > 1$) indicates worse performance.

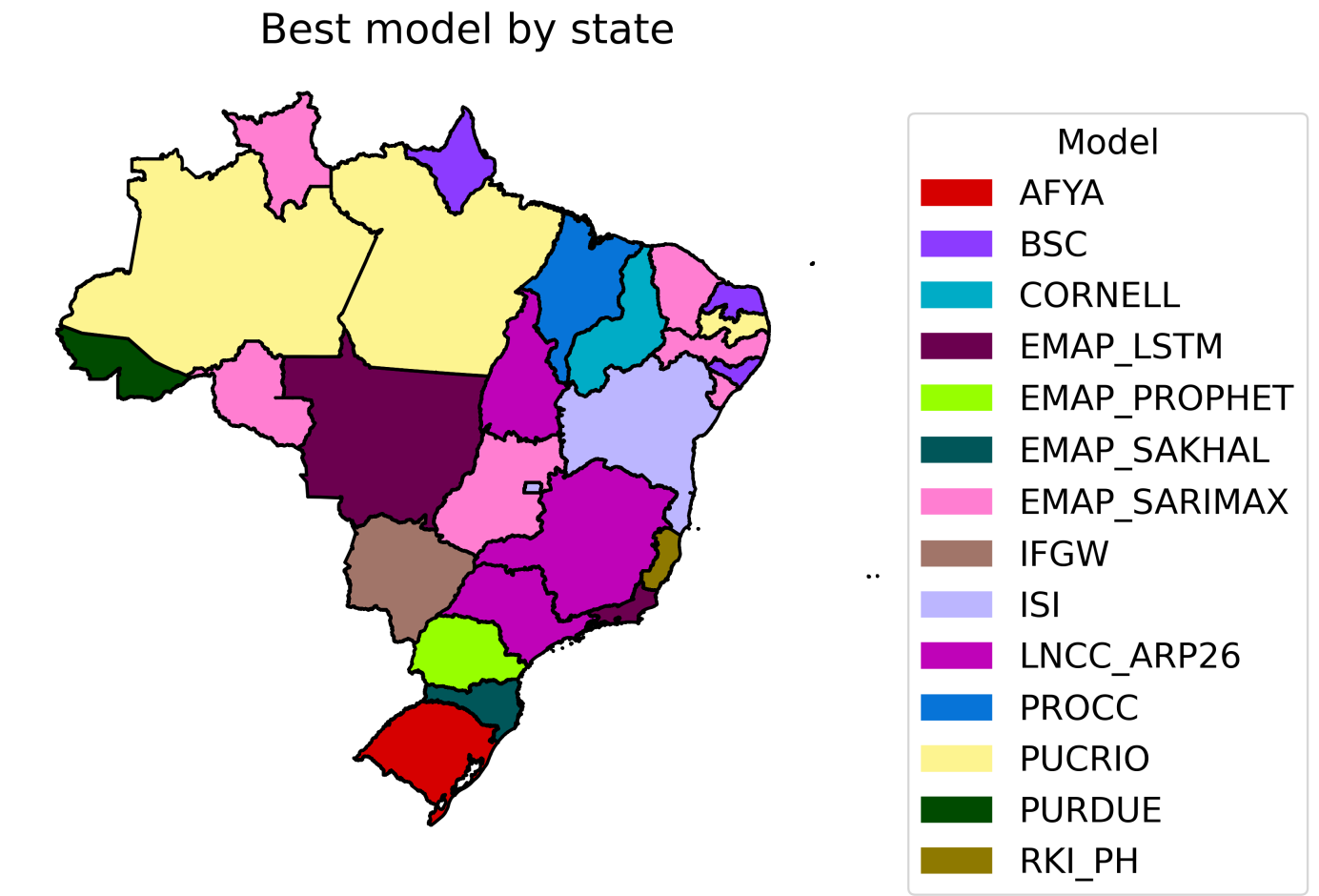
To obtain a state-level ranking, we computed the geometric mean of (R_v) across the four validation periods for each model. Models were then ranked within each state according to this metric, with lower values indicating better predictive performance.

Best-performing models per state

The bar plot below shows the number of states in which each model achieved the highest rank.



The map below displays the top-ranked model for each state. States are colored according to the model that achieved the best ranking.



The figures below present, for each region, the three highest-ranked models based on their performance in each state (using WIS^{norm}).

Medal board

South region:

MEDAL	PR	SC	RS
GOLD	EMAP_PROPHET	EMAP_SAKHAL	AFYA
SILVER	CERI	AFYA	EMAP_PROPHET
BRONZE	EMAP_XGB	UNESP	PUCRIO

Southeast region:

MEDAL	MG	RJ	SP
GOLD	LNCC_ARP26	EMAP_LSTM	LNCC_ARP26
SILVER	PUCRIO	RKI_LSTM	PUCRIO
BRONZE	BSC	PUCRIO	EMAP_SAKHAL

Midwest region:

MEDAL	DF	GO	MT	MS
GOLD	ISI	EMAP_SARIMAX	EMAP_LSTM	IFGW
SILVER	IFGW	EMAP_LSTM	BSC	LNCC_ARP26
BRONZE	RKI_LSTM	PROCC	EMAP_PROPHET	EMAP_SARIMAX

Northeast region:

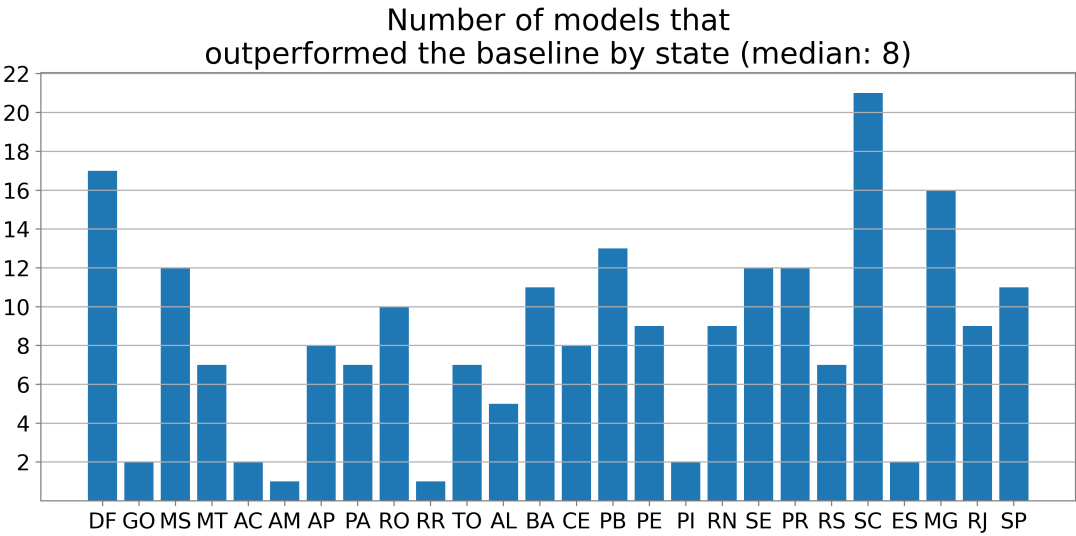
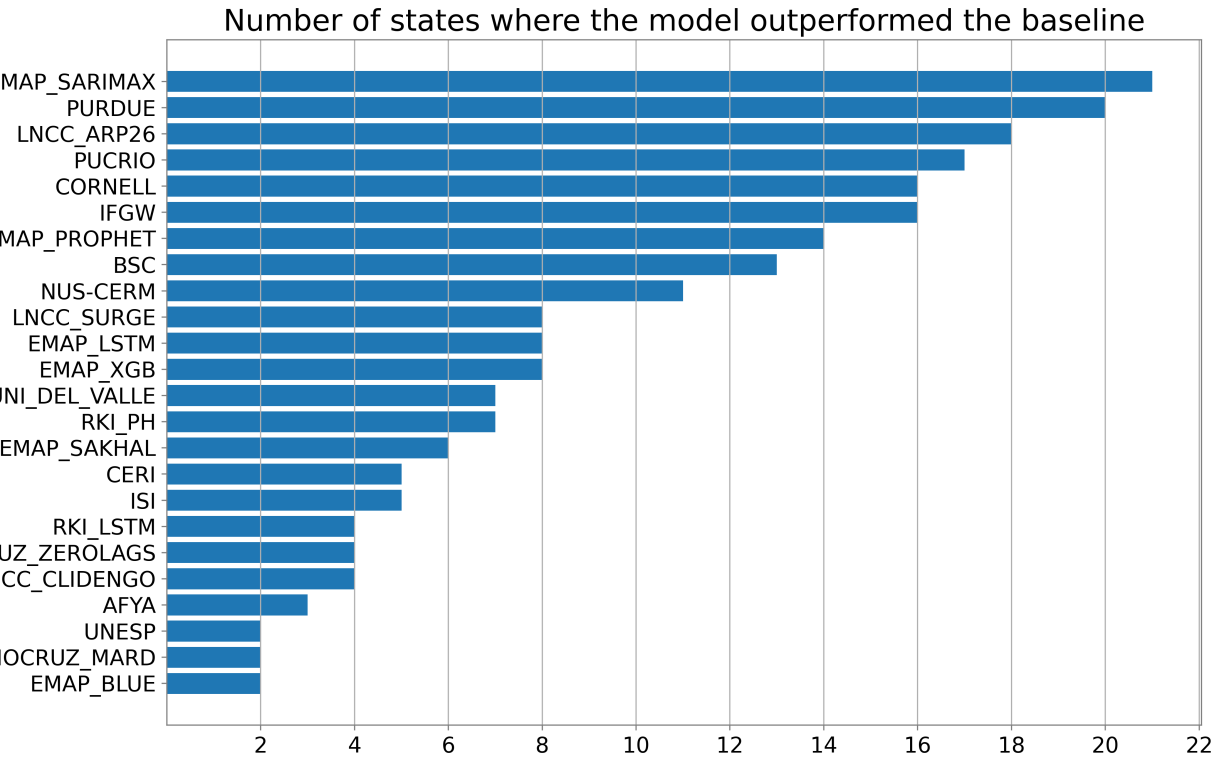
MEDAL	AL	BA	CE	MA	PB	PE	PI	RN	SE
GOLD	BSC	ISI	EMAP_SARIMAX	PROCC	PUCRIO	EMAP_SARIMAX	CORNELL	BSC	EMAP_SARIMAX
SILVER	EMAP_SARIMAX	IFGW	ISI	PUCRIO	EMAP_SARIMAX	PUCRIO	LNCC_ARP26	PUCRIO	LNCC_ARP26
BRONZE	IFGW	LNCC_ARP26	LNCC_ARP26	CORNELL	BSC	LNCC_ARP26	PROCC	EMAP_SARIMAX	IFGW

North region:

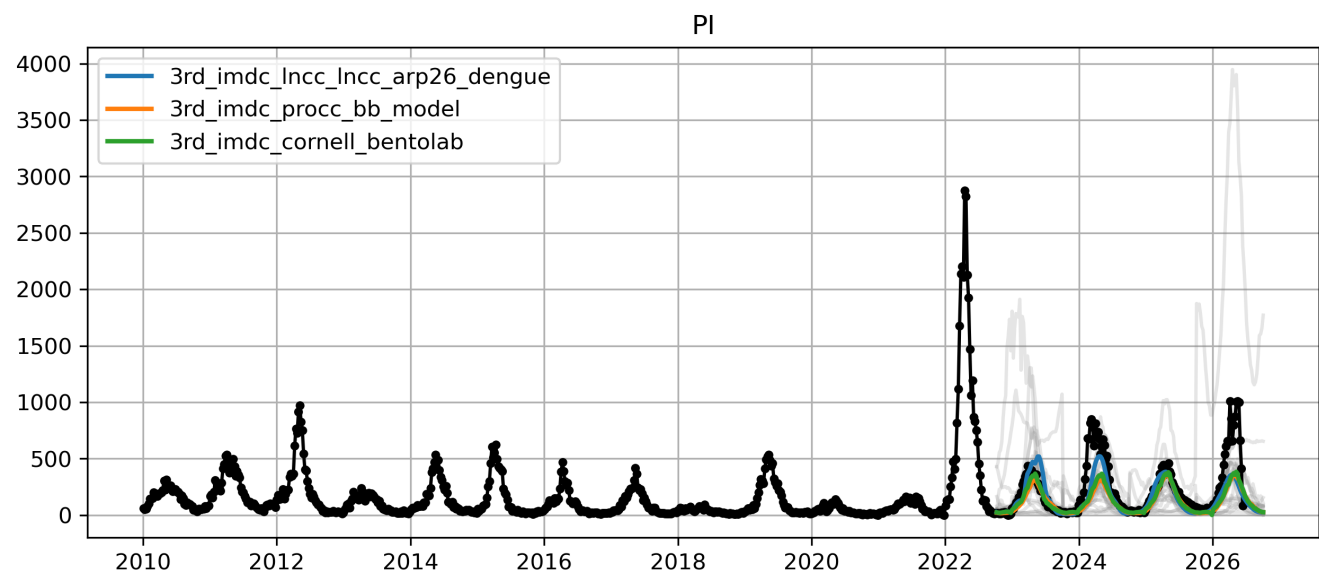
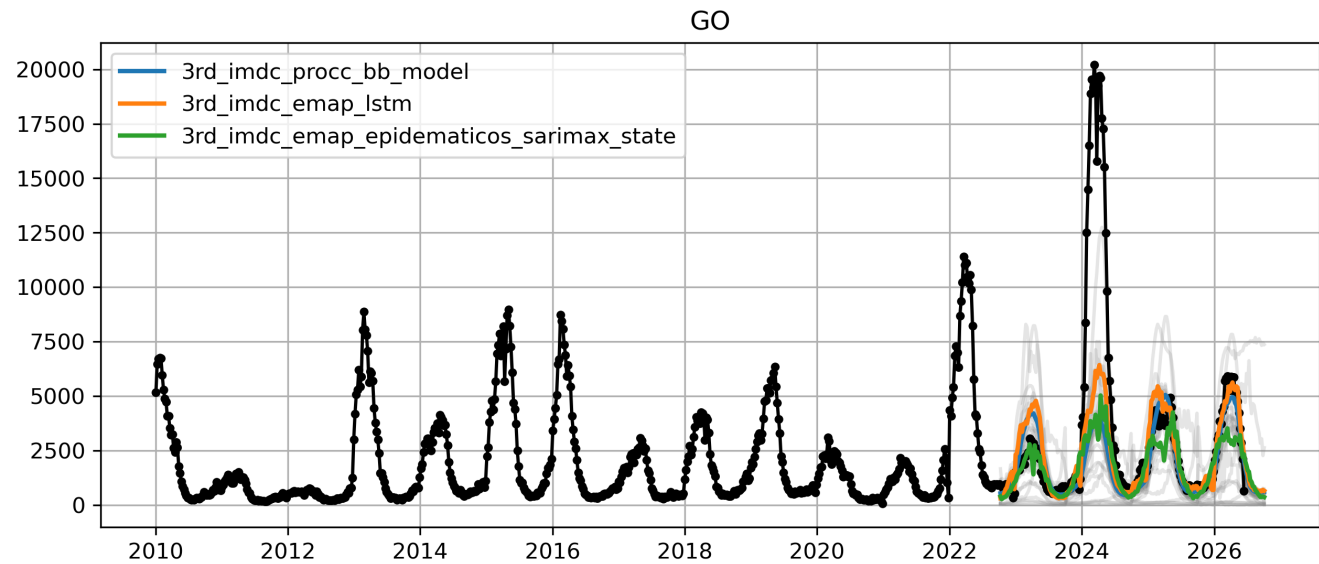
MEDAL	AC	AM	AP	PA	RO	RR	TO
GOLD	PURDUE	PUCRIO	BSC	PUCRIO	EMAP_SARIMAX	EMAP_SARIMAX	LNCC_ARP26
SILVER	PUCRIO	PROCC	EMAP_XGB	LNCC_ARP26	PUCRIO	PROCC	EMAP_SARIMAX
BRONZE	PROCC	IFGW	IFGW	PURDUE	LNCC_ARP26	IFGW	EMAP_LSTM

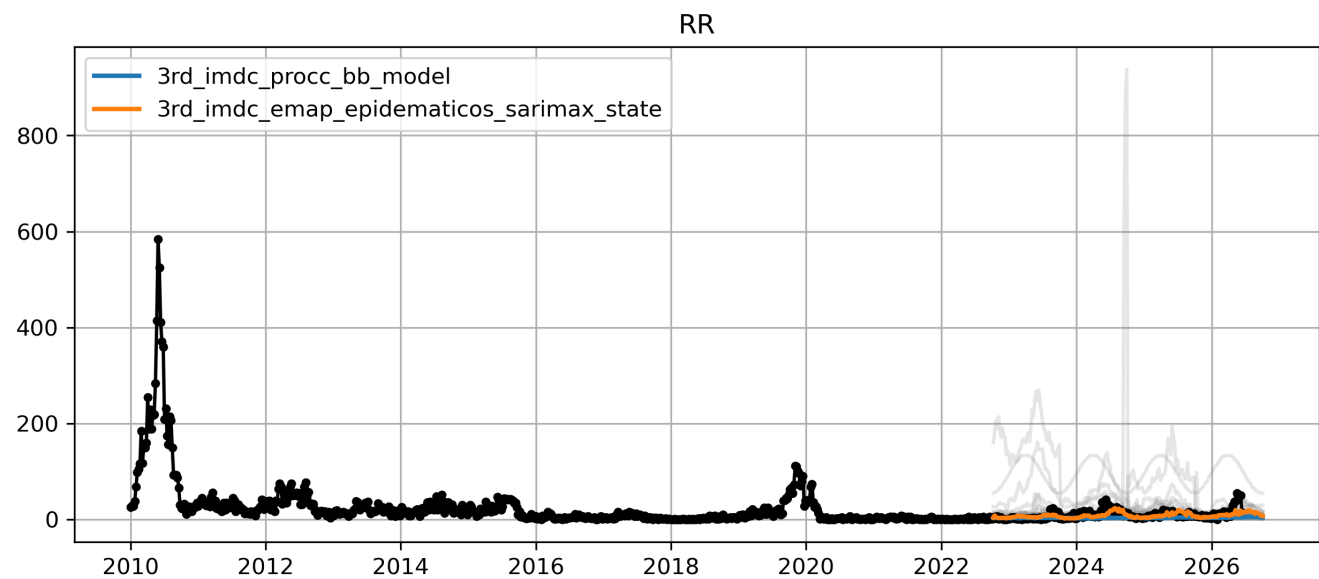
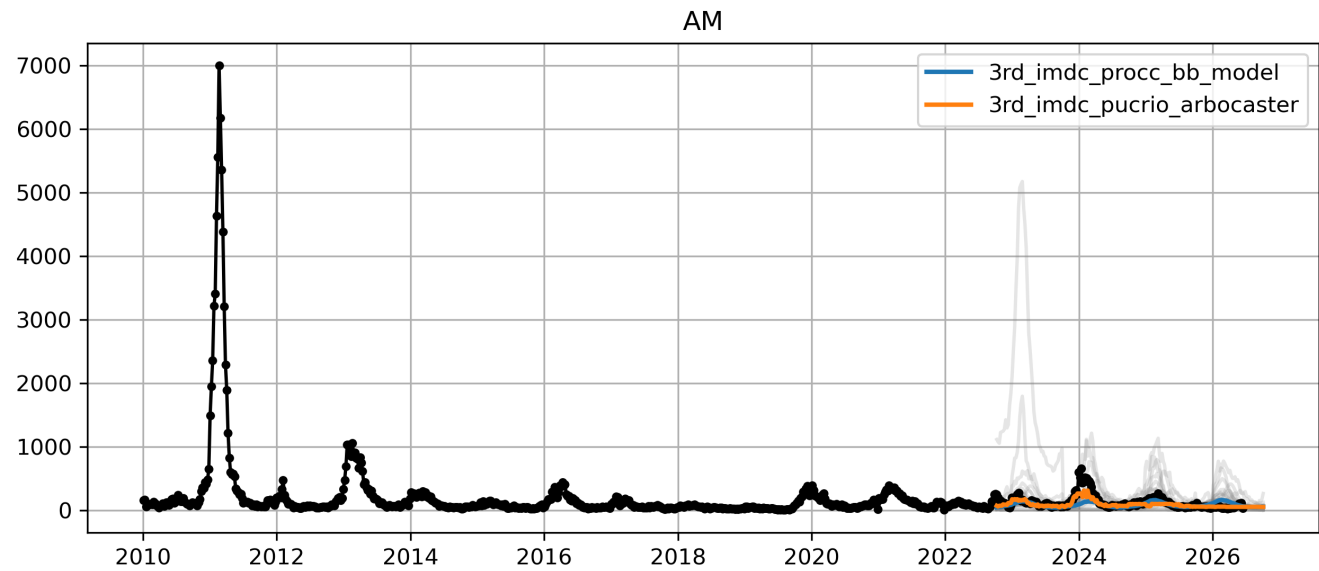
Overall model performance

Filtering by the models with mean \$R_v < 1\$ by state we generated the following plots:

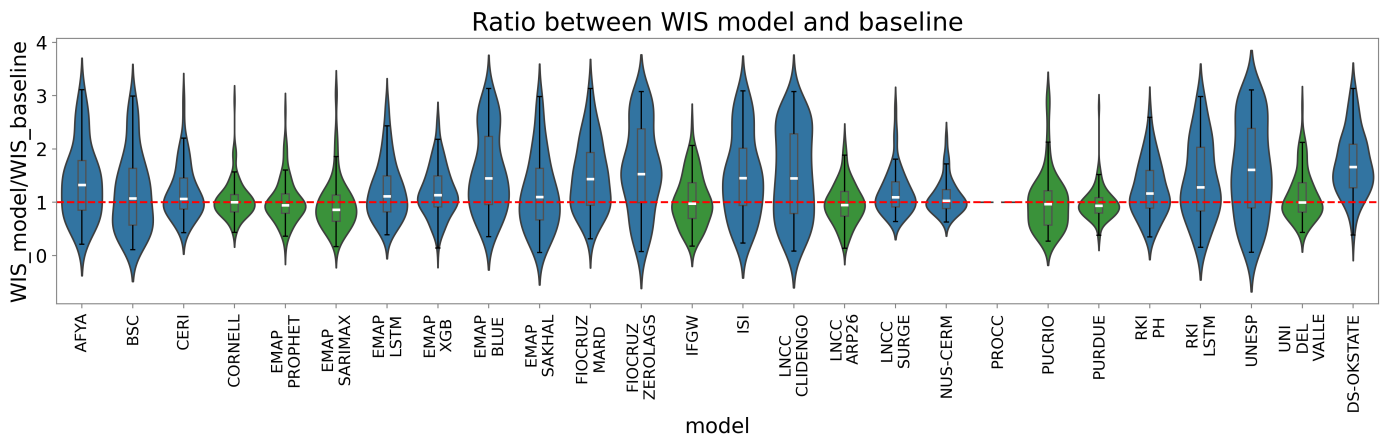


Zoomed-in view of the states with only a few models outperforming the baseline (AM, PI, GO, and RR). The baseline model is shown in blue, models that outperform the baseline are highlighted according to the legend, and all other models are shown in gray. The black lines represent the observed data:

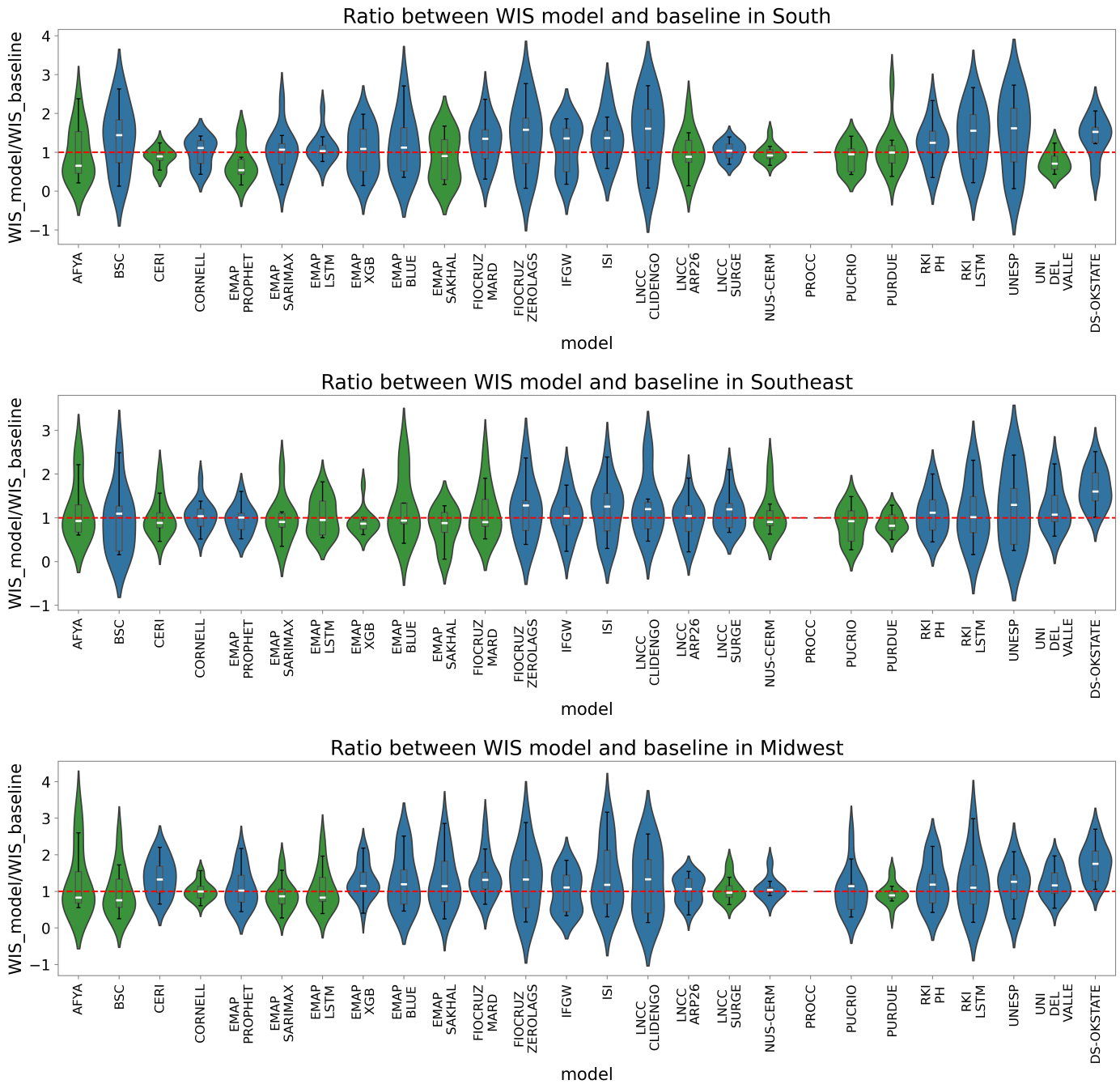


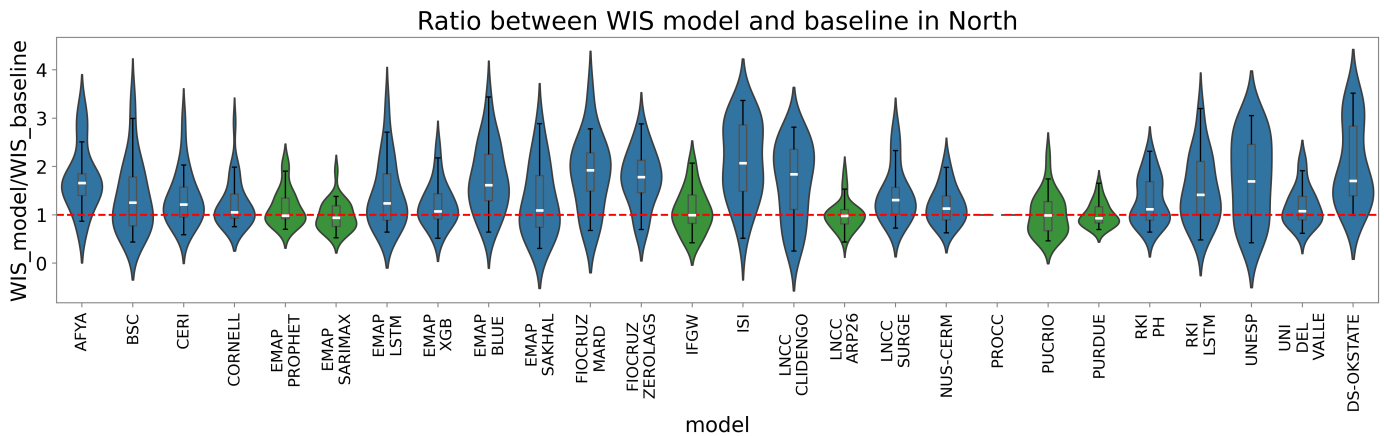
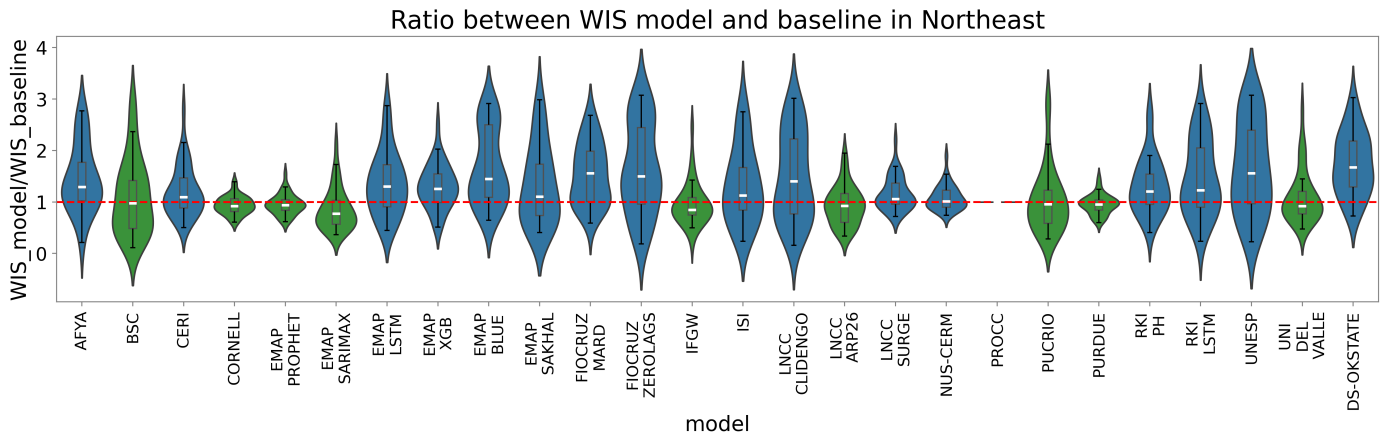


The violin plots below show the distribution of the performance ratio ((R)) across all validation periods. This analysis can be performed using data from all states, as shown below,



or restricted to a specific region, such as the Southeast:

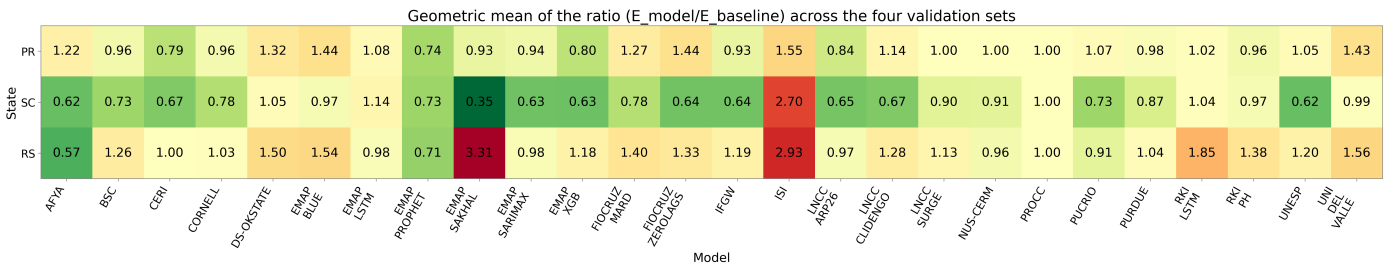




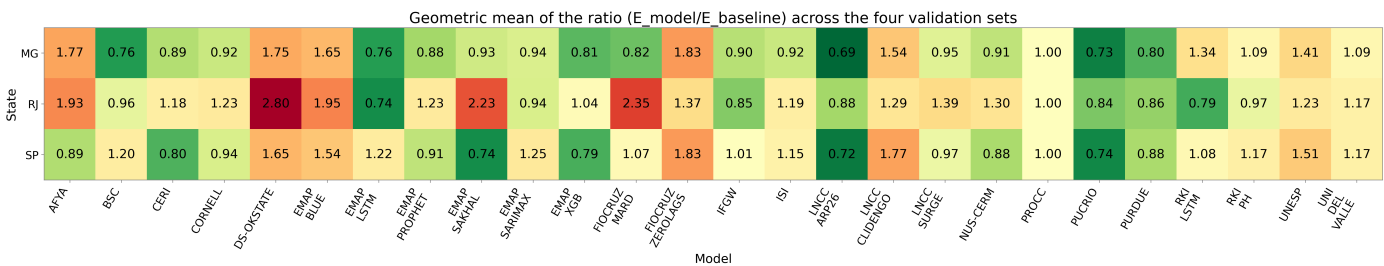
Heatmap by region

The figures below show the mean performance ratio for each state, grouped by Brazilian region.

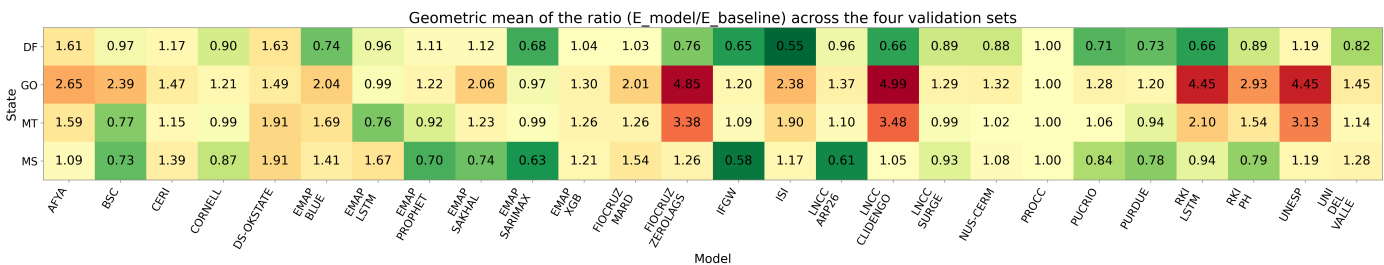
South region:



Southeast region:



Midwest region:



Northeast region:

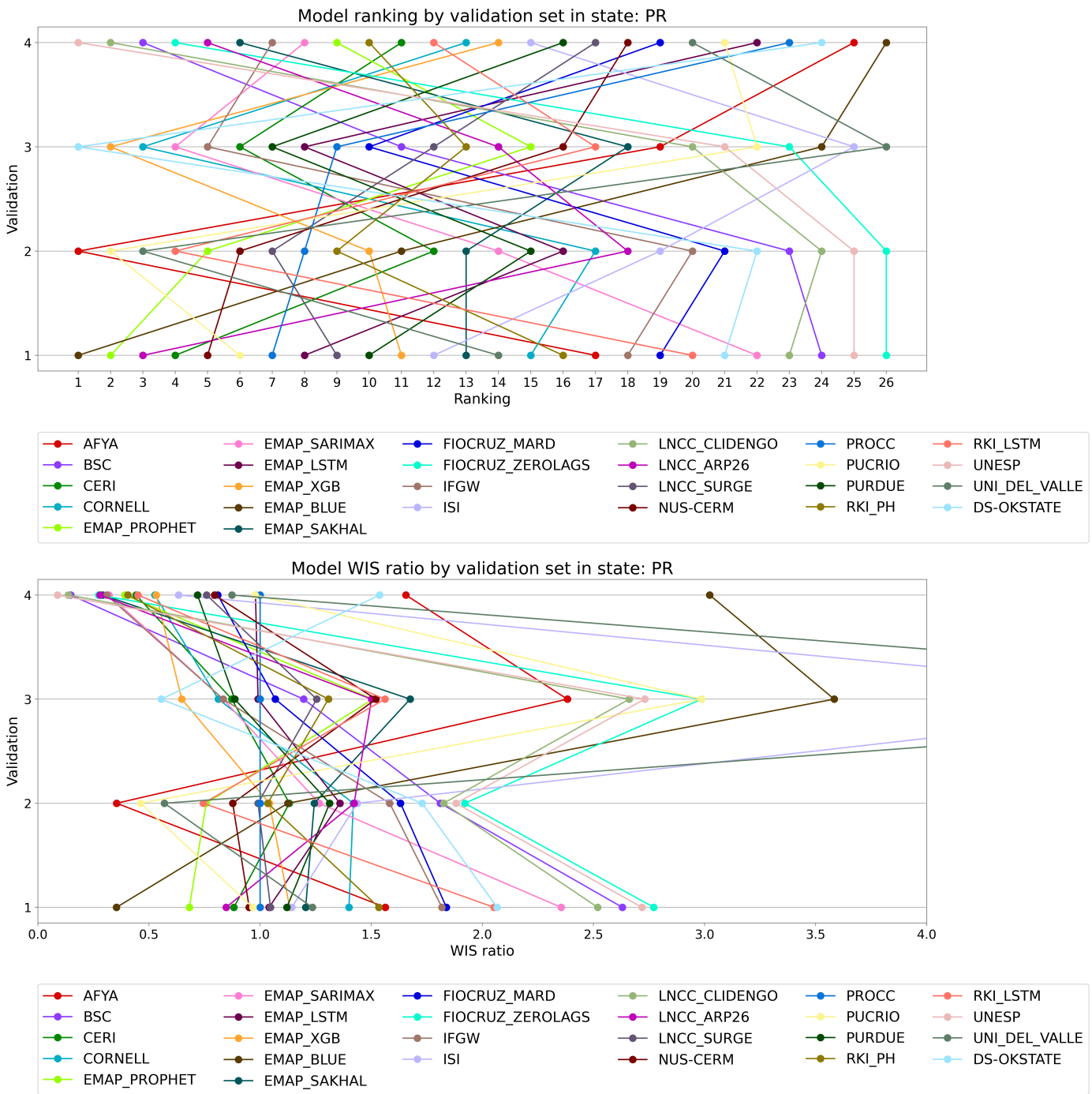
		Geometric mean of the ratio (E_model/E_baseline) across the four validation sets																											
State	AL	2.08	0.58	1.33	1.05	2.43	2.23	1.15	1.07	2.23	0.71	1.61	2.26	1.19	0.76	2.51	0.93	1.10	1.22	1.24	1.00	2.90	0.96	5.02	3.31	1.67	1.58		
	BA	1.80	1.16	1.08	0.90	1.50	2.47	0.90	0.86	2.43	0.84	1.02	1.08	2.15	0.79	0.71	0.81	1.59	1.07	0.96	1.00	0.99	0.82	1.21	1.18	1.65	0.91		
	CE	1.27	0.78	1.25	1.11	3.23	2.11	1.54	1.01	2.29	0.68	1.63	3.83	0.93	0.93	0.73	0.76	0.84	1.38	1.08	1.00	1.98	0.96	1.37	1.32	1.04	1.18		
	MA	1.23	1.49	1.07	1.02	1.66	1.64	2.23	1.06	7.24	1.20	1.17	1.96	2.12	1.31	3.39	1.49	2.79	1.07	1.23	1.00	1.00	1.12	2.32	1.80	2.92	1.20		
	PB	3.33	0.58	0.95	0.76	1.74	1.31	1.31	0.77	1.17	0.52	1.38	1.52	1.00	0.82	1.05	0.61	1.03	0.88	0.94	1.00	0.51	0.88	1.70	1.31	1.25	0.72		
	PE	1.35	1.40	1.10	0.91	1.75	1.52	1.65	0.96	4.80	0.70	0.97	1.75	1.97	0.78	1.24	0.74	1.61	1.13	1.00	1.00	0.74	0.97	1.59	1.30	1.95	0.78		
	PI	1.83	1.95	1.14	0.96	1.21	1.59	1.28	1.08	3.66	1.49	1.00	1.77	2.97	1.10	3.79	1.00	3.48	1.19	1.15	1.00	1.28	1.09	3.60	2.63	3.67	1.24		
	RN	3.20	0.56	1.15	0.91	2.20	1.50	1.42	0.91	0.83	0.63	1.56	2.16	1.06	0.84	1.43	1.01	0.97	1.17	1.07	1.00	0.60	0.88	2.10	1.54	1.24	1.18		
	SE	1.86	0.76	1.03	0.74	1.36	1.01	1.30	0.73	3.96	0.62	0.95	1.26	2.84	0.72	1.67	0.63	1.03	1.04	0.90	1.00	0.87	0.78	2.94	2.23	0.92	0.89		
		Model	AFYA	BSC	CERI	CORNELL	DS-OKSTATE	EMAP BLUE	EMAP LSTM	EMAP PROPHET	EMAP SAKPAL	EMAP SANNMAX	EMAP XGB	FIOCRUZ MARD	FIOCRUZ ZERO-LASS	IFGW	ISI	LNCC ARP2S	LNCC CLUIDENGO	LNCC SURGE	NUS-CERM	PROCC	PUCRIO	PURDUE	RKI LSTM	RKI PH	UNESP	UNI DEL VALLE	

North region:

		Geometric mean of the ratio (E_model/E_baseline) across the four validation sets																										
State	AC	1.80	1.01	1.24	1.21	2.34	1.89	1.72	1.11	1.22	1.18	1.06	3.66	1.62	1.01	2.59	1.11	1.07	1.74	1.29	1.00	0.97	0.94	1.63	1.35	2.30	1.06	
	AM	1.30	1.08	1.33	1.53	2.54	2.56	5.70	1.70	5.38	1.09	1.27	5.44	1.65	1.01	3.29	1.04	1.38	2.19	1.35	1.00	0.82	1.14	1.86	1.48	2.57	1.30	
	AP	2.33	0.73	1.33	0.93	1.62	1.60	1.48	1.00	1.40	0.93	0.81	1.64	3.38	0.90	0.99	1.04	1.23	1.24	0.96	1.00	4.40	0.93	1.17	1.10	1.32	1.27	
	PA	1.50	2.24	1.22	1.03	1.69	1.88	1.12	1.11	8.49	0.96	0.93	2.01	2.30	1.39	3.78	0.87	2.67	1.18	1.00	1.00	0.86	0.93	1.74	1.36	3.08	0.93	
	RO	1.93	1.16	1.45	0.96	2.08	2.08	1.26	0.92	3.77	0.77	1.32	1.57	1.95	0.85	2.16	0.80	1.31	1.03	1.00	1.00	0.79	0.91	0.90	0.93	1.21	1.03	
	RR	1.51	1.98	1.79	1.63	2.89	1.19	1.86	1.06	7.59	0.89	1.39	7.13	23.61	1.03	2.47	1.08	2.01	1.27	1.44	1.00	3.94	1.14	1.87	1.73	1.58	1.23	
	TO	3.28	1.13	1.19	1.01	2.22	1.25	0.86	0.98	1.28	0.75	1.27	3.00	1.37	0.88	1.86	0.75	1.24	1.02	1.12	1.00	0.90	0.89	2.14	1.65	1.53	1.13	
		Model	AFYA	BSC	CERI	CORNELL	DS-OKSTATE	EMAP BLUE	EMAP LSTM	EMAP PROPHET	EMAP SAKPAL	EMAP SANNMAX	EMAP XGB	FIOCRUZ MARD	FIOCRUZ ZERO-LASS	IFGW	ISI	LNCC ARP2S	LNCC CLUIDENGO	LNCC SURGE	NUS-CERM	PROCC	PUCRIO	PURDUE	RKI LSTM	RKI PH	UNESP	UNI DEL VALLE

Performance by state - all validations

To complement the analysis above, the following plots can be used to show the model performance across validation sets. The validation set is displayed on the y-axis, while either the ranking or the performance ratio for each validation period is shown on the x-axis. Each colored line represents a different model, allowing for a comparison of their performance consistency across validation periods.

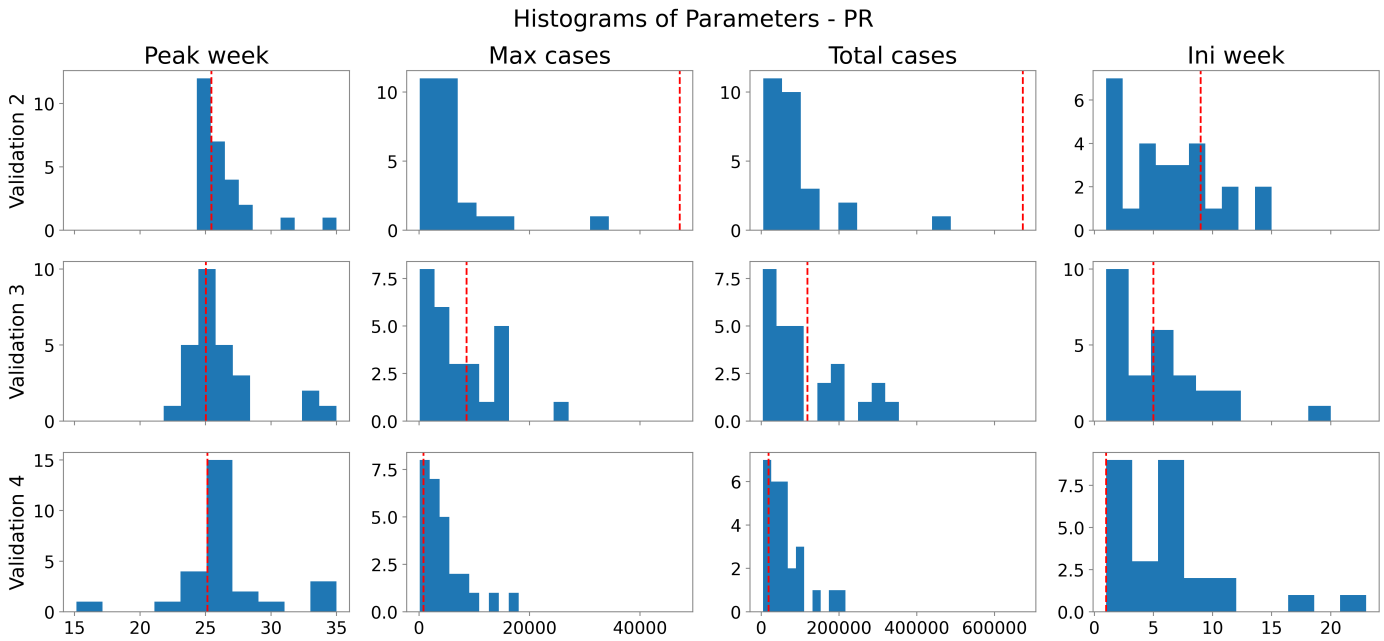


New metrics

Additionally, using the copula-based method described below, we transformed each model forecast into log-normal distribution parameters. We then generated 1,000 samples from these distributions and computed the total number of cases and the maximum number of cases, including their corresponding confidence intervals.

From the generated samples, we selected 100 samples and fitted the Richards model to estimate the peak week.

The plots below show the histogram distributions of each estimated parameter for each validation period. The red dashed line indicates the observed value from the data. The parameters are shown starting from validation 2 because one validation period is required to estimate the (ρ) parameter used in the copula method.



Bar plot of model parameters for a single state and validation set.

